

Surgical Chain-of-Thought Reasoning in Frontier Large Language Models

Elizabeth Schaefer
elizabeth.schaefer@yale.edu

Advisor: Arman Cohan
arman.cohan@yale.edu

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Abstract

Large language models (LLMs) are frequently being used for clinical decision support, yet their usefulness in surgery depends not only on the accuracy of their recommendations but also on the transparency and reliability of their reasoning. This thesis presents a multi-domain evaluation pipeline for surgical chain-of-thought reasoning built on de-identified discharge summaries and associated notes from MIMIC-IV-Note v2.2. Surgical cohorts for inguinal hernia repair, head and neck cancer, bariatric surgery, surgical oncology, and aortic surgery are constructed using curated ICD-9/10 diagnosis and procedure code lists. For each cohort, temporally-ordered note chains are formatted into structured prompts that elicit step-by-step clinical reasoning from subsets of five frontier LLMs.¹ The resulting model outputs are then evaluated with a five-dimensional rubric that scores clinical knowledge application, contextual understanding, hallucination control, reasoning transparency, and chain-of-thought quality. To explore scalable evaluation, this thesis further tests a frontier LLM (gpt-4o) as a “meta-judge” of inguinal hernia reasoning. The LLM judge achieves 58-75% exact agreement and 91-96% agreement within ± 1 point of surgeon reference scores across all dimensions, while being slightly more lenient on average. Together, these results showcase the strengths and limitations of current LLM chain-of-thought reasoning in surgical contexts, and demonstrate how calibrated LLM judges could support the future scaling of expert-informed evaluation.

¹GPT-5-Mini, GPT-5-Chat, Gemini 2.5 Pro, Claude 4.1 Opus, DeepSeek-R1, and Llama-3.1.

1 Introduction and Related Work

Advancements in large language models (LLMs) have opened many new possibilities for assisting medical decision making, from answering textbook questions to summarizing clinical notes. However, real-world patient care, especially in intense surgical settings, needs more than just quick answers. Surgeons must translate and integrate longitudinal histories, imaging, comorbidities, and hospital guidelines into transparent and justifiable plans. Any model intended to assist this process must therefore not only provide accurate recommendations, but also clear reasoning that clinicians can scrutinize, contest, and ultimately trust.

Chain-of-thought reasoning provides one mechanism for making this process visible. Chain-of thought is the natural-language reasoning trace an LLM produces between input and decision; making this trace explicit, rather than jumping immediately from intaking Question A to outputting Answer B, has been shown to help on challenging reasoning benchmarks [1]. In a randomized trial, clinicians with access to an LLM achieved higher diagnostic accuracy on clinical vignettes [2], while MedReason [3] showed that converting clinical QA into knowledge-graph “thinking paths” and training on these chain-of-thoughts improves problem-solving. Together, these results suggest that such chain-of-thought reasoning is valuable for medical decision support.

At the same time, frontier models are at a crossroads in emphasizing human-interpretable reasoning. While DeepSeek-R1-Zero has demonstrated superior performance on mathematical QA benchmarks, the chain-of-thought outputs mix languages and are less interpretable, and enforcing more language-consistent chain-of-thought degrades benchmark accuracy [4]. This represents a worrisome development where commercial models must choose between performance gains and a comprehensible thinking process. OpenAI espouses a firm stance on prioritizing the latter [5], noting that such monitoring may be one of humanity’s few tools to supervise powerful models; they further caution that naïvely penalizing “bad” thoughts can teach models to hide their true reasoning rather than change it.

I share this emphasis on transparency: even if insisting on comprehensible CoT degrades some quantitative performance metrics, it is indispensable in clinical settings where clinicians need to see how a recommendation was made. This project therefore asks: to what extent do current frontier LLMs demonstrate clinically usable chain-of-thought reasoning on structured surgical decision-making tasks? To answer this, I construct a multi-domain evaluation pipeline that applies LLMs to longitudinal surgical note chains from MIMIC-IV (including inguinal hernia, head and neck cancer, bariatric surgery, and aortic repair), elicits their reasoning, and annotates this reasoning using expert physician scoring. However, evaluating this reasoning at scale is still difficult: recent medical benchmarks, such as OpenAI HealthBench [6] and MedAgentBench [7], show that rig-

orous evaluation (not just new models) is now the limiting step for clinical LLMs. I also experiment with a rubric-guided LLM judge to assess scaling capabilities, or “meta-evaluation,” in hopes of eventually expanding the annotated corpus using a hybrid human–LLM–judge. In sum, this thesis focuses on evaluating the chain-of-thought reasoning of large language models in surgical decision-making settings.

2 Methodology

2.0.1 Data Source and Cohort Construction

I used de-identified discharge summaries and other associated notes from the MIMIC-IV-Note v2.2 database. For this thesis, I constructed three surgical cohorts: (1) inguinal hernia repair (GI & endoscopic surgery), (2) head and neck cancer, (3) a mini-cohort comprising bariatric surgery, surgical oncology, and aortic surgery. Within each domain, I identified candidate patients using curated ICD-9/10 diagnosis and procedure code lists (A.2). For each included patient, all relevant notes were extracted and sorted chronologically by subject ID and chart time. After filtering, the combined dataset contained approximately 12,000 unique patients and over 80,000 notes across the specialties.

2.0.2 Note Formatting and Prompting

To explore different aspects of clinical reasoning, I developed three prompting settings:

- Single-note reasoning: a single discharge summary is provided and the model is asked to justify a specific surgical decision.
- Longitudinal chain reasoning: multiple notes from a single patient are aligned in temporal order by charttime, and the model must integrate information across time to answer the prompt.
- Concise answer generation, where the model produces a short, decision-focused answer after reasoning.

In each setting, notes were wrapped in a consistent template indicating the surgical domain and a related decision-making task (e.g., “summarize the surgery,” “identify a plausible alternative surgery,” “explain the rationale for choosing the documented approach over the alternative”). These prompts were often domain-specific and can be viewed in the Appendix.

2.0.3 Model Infrastructure

To meet the privacy and reproducibility requirements of MIMIC, I migrated the entire pipeline to offline API environments for GPT-5, Gemini 2.5 Pro, Claude 4.1 Opus, DeepSeek-R1, and Llama-3.1. A command-line interface enables the full workflow of specifying ICD-based inclusion and exclusion criteria, selecting domains, batching notes, and automatically generating model completions under different prompts and requirements.

```
○ ejs93@lambda1:~$ python pipeline.py --mode completions \  
  --icd_version 9 --inclusion_icd 550 \  
  --platform vertexai --model gemini-2.5-pro \  
  --completion_count 10 --seed 1 --temperature 0.2 --max-tokens 512
```

Figure 2.1: Example of command line completions

2.0.4 Annotation Rubric and Expert Evaluation

I worked with a total of eight physicians (many of whom were professors of surgery) to define a rubric for evaluating model outputs along five dimensions: (1) clinical knowledge application, (2) contextual understanding, (3) hallucination control, (4) reasoning transparency, and (5) chain-of-thought quality. The full rubric text and scoring guides are provided in the Appendix (A.1).

For the inguinal hernia cohort, four physicians independently scored each model’s response for every case. For the head and neck cancer cohort, five physicians evaluated the outputs of a single model. For models that expose both raw and prompted reasoning traces, annotators scored both; for models without raw CoT access, only prompted reasoning was evaluated.

2.0.5 LLM Judge Methodology

In addition to these human annotations, I experimented whether an LLM could serve as a “meta-judge” of surgical reasoning on the inguinal hernia cohort. To build a reference standard, I started from the full inguinal hernia set of surgeon annotations and treated each evaluated case-model pair as one item. I then measured how often each pair of surgeons gave the same scores on the same items and identified the two surgeons who agreed the most with one another on a reasonably large overlap of cases (70%). For that best-agreeing pair, I averaged their scores to create a single “gold” label per item, which I treated as the human reference for comparison.

I then used a frontier LLM (gpt-4o) as a judge. For each item, the judge read the patient note, the clinical question, the model’s reasoning, and its final answer, and produced 1-5 scores on the same annotation rubric. The prompts were engineered to slightly encourage the judge towards behaving like a careful surgeon rather than a perfectionist grader (for example, not punishing concise but clear reasoning). Finally, I compared the judge’s scores to the gold surgeon labels by looking at mean score differences and how often they exactly matched or fell within one point of each other for each dimension.

3 Results

3.0.1 Formal Annotation Corpus

Inguinal Hernia Cohort (Gastrointestinal & Endoscopic Surgery)

I generated a cohort of inguinal hernia discharge summaries using diagnosis and procedure-based ICD filters (A.2.2), from which a professor of surgery then selected 24 representative cases. Each case was paired with a complex, surgery-grounded prompt asking each generative model to summarize the operation, consider alternatives, and justify the documented approach (full prompt in A.4.2).

I evaluated six models in this setting: GPT-5-Mini, GPT-5-Chat, DeepSeek-R1, Claude 4.1 Opus, Gemini 2.5 Pro, and Llama-3.3-70B. For models that expose both raw and prompted reasoning traces (Claude, Gemini, DeepSeek), annotators scored:

1. Raw reasoning transparency and chain-of-thought quality, and
2. Prompted reasoning transparency and chain-of-thought quality.

For models without raw CoT access (GPT-5-Mini, GPT-5-Chat, Llama-3.3-70B), only prompted reasoning was scored. Four physicians independently scored each model output on the five rubric dimensions.

Head and Neck Cancer Cohort (Oncology)

I generated a cohort of head and neck cancer discharge summaries using diagnosis and procedure-based ICD filters (A.1), from which a professor of surgery then selected 20 representative cases. Each case was paired with an oncologic decision-making prompt focused on surgical intervention (full prompt in A.4.1).

In this setting, a single LLM (Llama-3-8B) was used to generate prompted chain-of-thought reasoning and final answers for all cases. Five physicians independently scored each output on the five rubric dimensions.

3.0.2 Illustrative Cross-Specialty Overview

Using the same methodology, I also constructed a small qualitative cohort to compare model behavior across three different specialties. I focused on the three domains of bariatric surgery, head and neck cancer, and aortic surgery to give a diverse sampling of clinical specialties. For each specialty, I first identified candidate patients from the filtered MIMIC-IV cohorts using the relevant ICD procedure codes. For every subject ID,

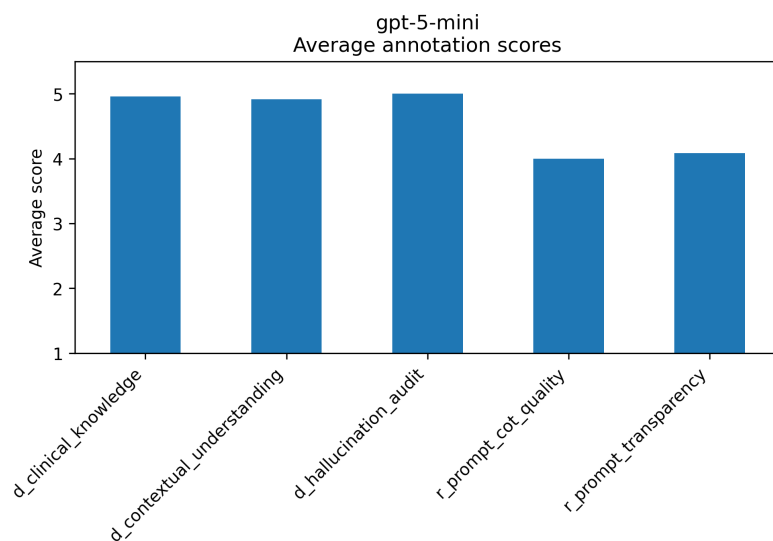


Figure 3.1: Average annotation scores for GPT-5-mini on inguinal hernia cohort

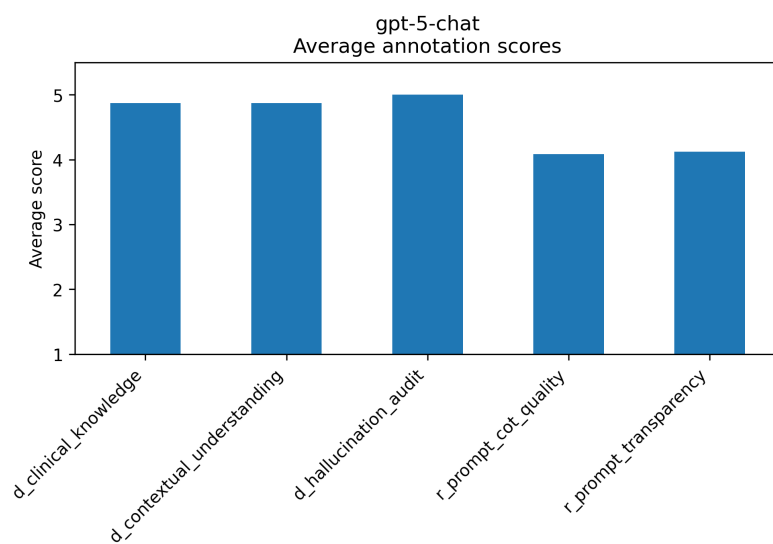


Figure 3.2: Average annotation scores for GPT-5-Chat on inguinal hernia cohort

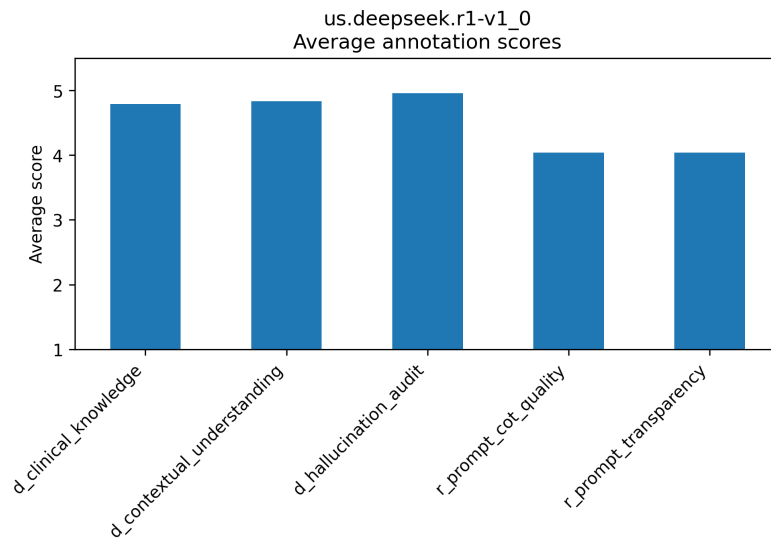


Figure 3.3: Average annotation scores for DeepSeek-R1 on inguinal hernia cohort

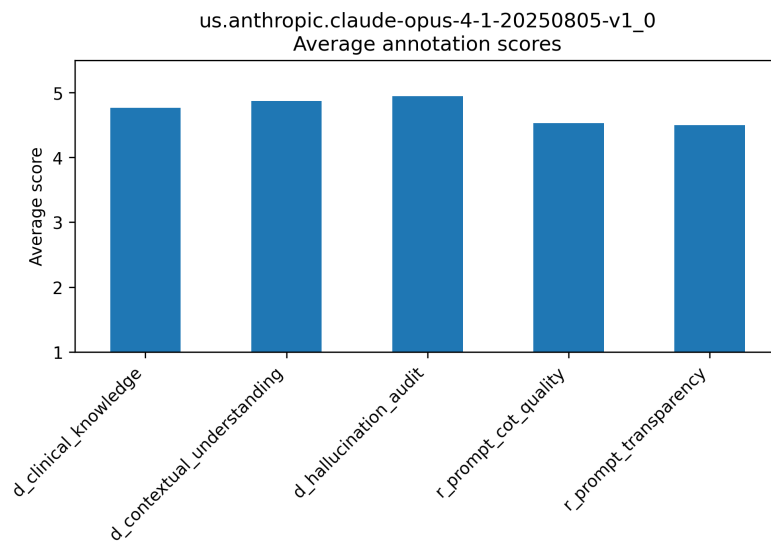


Figure 3.4: Average annotation scores for Claude Opus 4.1 on inguinal hernia cohort

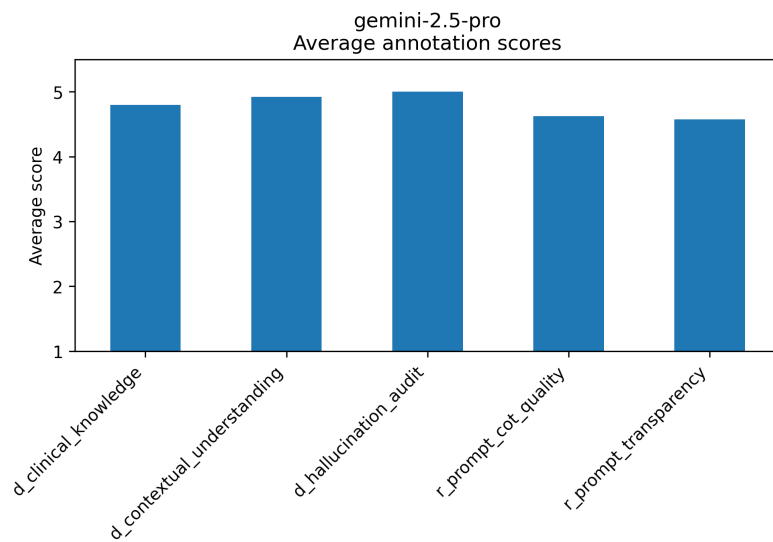


Figure 3.5: Average annotation scores for Gemini 2.5 Pro on inguinal hernia cohort

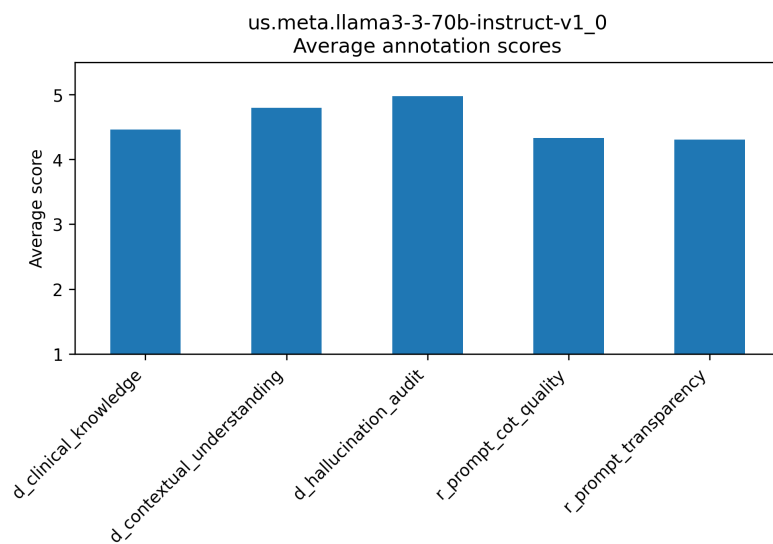


Figure 3.6: Average annotation scores for Llama-3.3-70B on inguinal hernia cohort

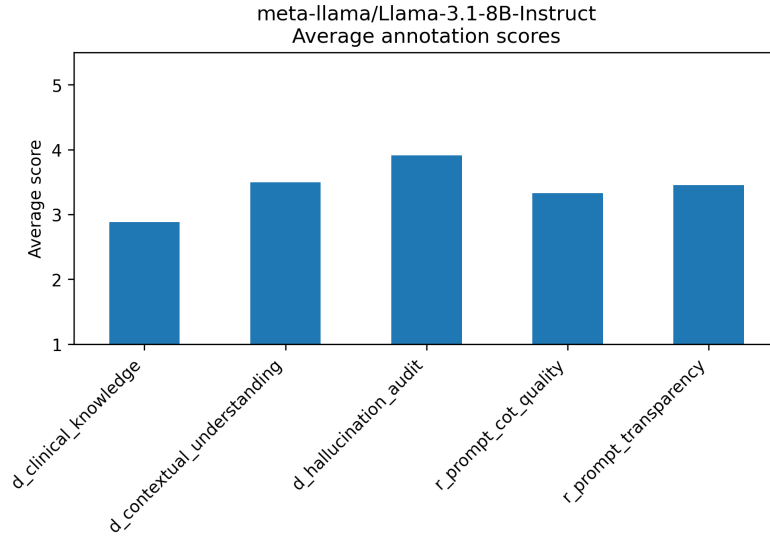


Figure 3.7: Average annotation scores for Llama-3.1-8B on head and neck cancer cohort

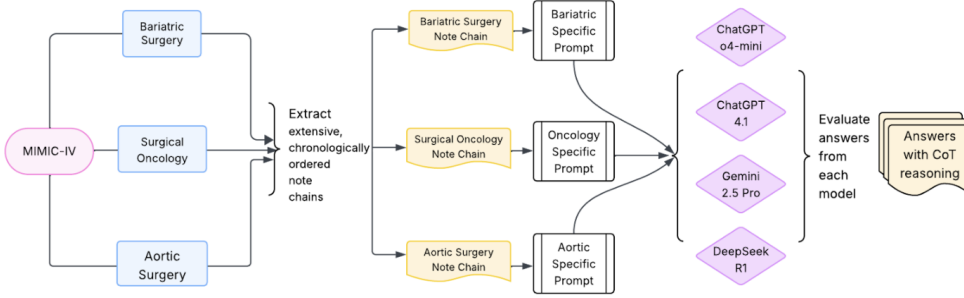


Figure 3.8: Workflow Overview

I aggregated discharge summaries and radiology reports and sorted all notes in chronological order by chart time. I then ranked the top-10 patients within each specialty by the length of their longitudinal note chain to prioritize the most well-documented admissions. The professor of surgery then selected two clinically representative patients from each top-10 set, yielding a final qualitative cohort of six patients (two bariatric, two head and neck, two aortic). These cases can be viewed in A.3.

Four LLMs (GPT-o4-mini, GPT-4.1, Gemini 2.5 Pro, and DeepSeek-R1) were then prompted with fixed, domain-specific tasks (A.4.3). This design yielded 24 outputs (6 patients $\times$ 4 models). The professor of surgery scored these responses on the five-dimension rubric, as seen in Figures 3.9 and 3.10.

3.0.3 LLM Judge Test

On the inguinal hernia cohort (24 cases), the LLM judge stayed very close to the surgeon reference ratings. Across all five dimensions, exact agreement ranged from 58-75%, and 91-96% of scores fell within ± 1 point of the gold labels. The judge was slightly more

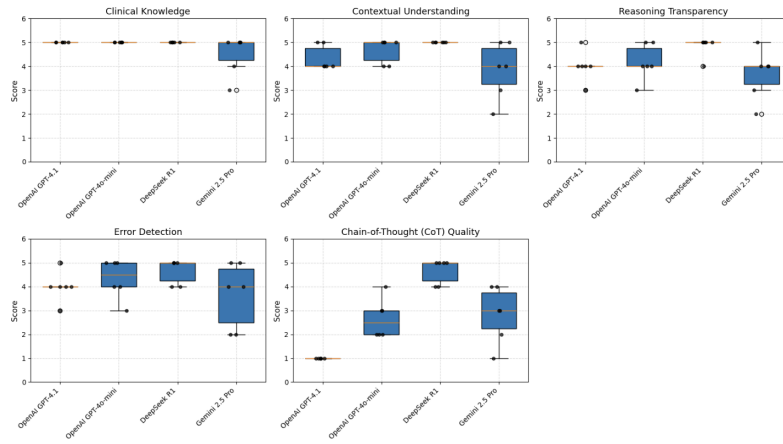


Figure 3.9: Comparing Performance by Model

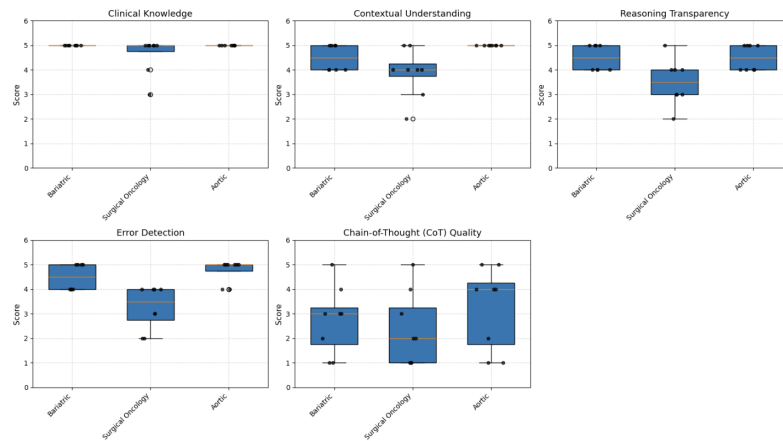


Figure 3.10: Comparing Performance by Domain

generous than surgeons (mean differences of about 0.2-0.3 points on a 1-5 scale). These results provide an initial estimate of how reliable LLM judges are in this setting, and also help inform whether future scaling of the pipeline can safely rely on some partial automation of the annotation process.

Metric	Exact match (%)	Within ± 1 (%)	Mean diff [†]	Corr
Reasoning transparency	62.5	91.7	-0.19	0.39
CoT quality	66.7	95.8	-0.19	0.46
Clinical knowledge	58.3	91.7	-0.29	–
Contextual understanding	66.7	95.8	-0.21	0.13
Hallucination audit	75.0	95.8	-0.19	–

Table 3.1: Agreement between LLM judge and surgeon gold labels on the inguinal hernia cohort (N = 24). [†]Mean diff = Judge - Gold; negative values indicate slightly higher judge scores.

4 Conclusions and Future Work

4.1 Conclusions

This thesis set out to evaluate how current frontier large language models reason about actual surgical cases, and to do so in a way that cares about transparency rather than only final answers. To that end, I constructed a multi-domain pipeline that intakes discharge summaries and associated notes from MIMIC-IV-Note v2.2, filters patients into different cohorts using domain-specific ICD-9/10 code lists, and formats temporally ordered note chains into structured prompts. These prompts elicit chain-of-thought reasoning and concise, decision-focused answers from multiple LLMs.

A central contribution of this work is a formal annotation corpus of surgical reasoning traces. Eight physicians, many of them professors of surgery, scored model outputs on a rubric spanning clinical knowledge application, contextual understanding, hallucination control, reasoning transparency, and chain-of-thought quality. This structured rubric makes it possible to observe patterns that would be invisible from final answers alone.

Finally, this thesis explores whether evaluation itself can be partially automated. Using gpt-4o as a meta-judge on the inguinal hernia cohort, I compare its rubric scores to a derived gold standard constructed from the most consistently agreeing annotators. The judge matches surgeon scores exactly on 58–75% of items across dimensions and falls within ± 1 point on 91–96% of items, while overall tending to be slightly more lenient. These findings suggest that calibrated LLM judges can approximate expert behavior closely enough to support scaling, but they are not yet interchangeable with human review, especially in safety-critical settings like surgery.

Taken together, this work contributes (1) a reusable pipeline for structured surgical reasoning evaluation on MIMIC-IV, (2) a surgeon-annotated corpus of chain-of-thought traces across multiple surgical domains, and (3) an initial assessment of LLM-based meta-judging for clinical reasoning. More broadly, my thesis argues that any serious deployment of LLMs in surgery must treat the quality and transparency of intermediate reasoning as prioritized objects of study, not just side effects of producing an answer.

4.2 Limitations and Future Work

Although this thesis demonstrates a practical framework for evaluating surgical chain-of-thought reasoning, it also has important limitations. All this data is drawn from a single source (MIMIC-IV-Note v2.2), and the cohorts cover a limited set of procedures and institutions. The annotation rubric, though developed in collaboration with surgeons, reflects one particular decomposition of “good” reasoning, and the final scores may be sensitive

to how individual annotators interpret its dimensions. The LLM judge experiments are restricted to a single model, cohort, and rubric, and therefore also provide only an initial indication of when such automated evaluation might be reliable.

I have several future directions in which I'd like to take this work. I have recently received IRB approval for YNNH data, which would be an amazing way to locally focus this project, but would also necessitate careful model privacy and patient confidentiality measures. On the modeling side, the current study treats LLMs as fixed black boxes and focuses on prompting and evaluation. A natural next step is to use the collected annotations and reasoning traces as supervision: for example, fine-tuning models or implementing a chain-of-thought reward function. Similarly, the prompts used here are relatively simple, and future work could compare more advanced prompting strategies and range of questions. I'd also like to continue experimenting with the LLM judge to further determine scalability fit, which would give us a wider cohort of annotations to train on and learn from. Another direction in meta-evaluation is to explore hybrid evaluation, where human experts focus on the most uncertain or contested cases while LLM judges handle routine examples, reducing annotation burden without sacrificing oversight.

Finally, this project has not directly addressed questions of equity, trust, or workflow integration. Future work could also examine how model reasoning quality varies across patient sub-groups, how surgeons interact with and override LLM chains-of-thought in simulated or real clinical scenarios, and what kinds of explanations lead to appropriate reliance rather than over-trusting. In the long term, the goal is not simply to make models better at reasoning like surgeons, but to build systems whose reasoning is robust, inspectable, and aligned with the values and constraints of real surgical practice. This thesis is a first step toward that goal by making surgical reasoning visible and measurable.

4.3 Resulting Publications and Patent

In parallel with this thesis, these methods and findings (in collaboration with the Yale NLP lab and the Yale School of Medicine) have already translated into external impact and recognition.

- **Schaefer, E.**, Li, X., Wang, B., Ghiassi, S., Gibbs, K.E., Pai, S.I., Vallabhajosyula, P., Krumholz, H., Schneider, E.B., Cohan, A., Ong, C.S., “Enhancing Complex Surgical Decision-Making with State-of-the-Art Reasoning Models,” 2025. Accepted for oral presentation at the *21st Annual Academic Surgical Congress (ASC)*, February 2026. To be presented by **Schaefer, E.**
- Lee, Y., **Schaefer, E.**, Wang, B., Li, X., Butensky, S., Flom, E., Zhou, R., Duffy, A., Gibbs, K., Schneider, E., Cohan, A., Ong, C.S., “Assessing the Reasoning Capabilities of State-of-the-Art Large Language Models in Complex Surgical Decision-Making Following Admission for Inguinal Hernia Repair,” 2025. Accepted at *SAGES*, March 2026. To be presented by Lee, Y.
- Aguirre, N., Wang, B., **Schaefer, E.**, Huang, J., Li, X., Kolamuri, S.R., Ying, L., Li, X., Ang, S.S., Schneider, E.B., Pai, S.I., Cohan, A., Ong, C.S., “Hippocrates-

Karkinos-o1: An Enhanced Model for Oncology Decision Support and Guideline-Aware Reasoning with Self-Refinement and Guideline-Informed Retrieval,” 2025. Under review at *NCCN* 2026.

- Ong, C.S., Cohan, A., Wang, B., **Schaefer, E.**, Schneider E. “System and Method for Clinical Reasoning,” 2025. *U.S. Provisional Patent, No. 63,932,193*.

Bibliography

- [1] Jason Wei et al. “Chain-of-Thought Prompting Elicits Reasoning in Large Language Models”. In: *Advances in Neural Information Processing Systems*. Vol. 35. 2022, pp. 24824–24837.
- [2] Ethan Goh, Robert Gallo, Jason Hom, et al. “Large Language Model Influence on Diagnostic Reasoning: A Randomized Clinical Trial”. In: *JAMA Network Open* 7.10 (2024), e2440969. DOI: 10.1001/jamanetworkopen.2024.40969.
- [3] Juncheng Wu et al. “MedReason: Eliciting Factual Medical Reasoning Steps in LLMs via Knowledge Graphs”. In: *arXiv preprint* (2025). arXiv: 2504.00993 [cs.CL].
- [4] Daya Guo, Dejian Yang, Haowei Zhang, et al. “DeepSeek-R1 Incentivizes Reasoning in LLMs through Reinforcement Learning”. In: *Nature* 645.8081 (2025), pp. 633–638. DOI: 10.1038/s41586-025-09422-z.
- [5] Bowen Baker et al. “Monitoring Reasoning Models for Misbehavior and the Risks of Promoting Obfuscation”. In: *arXiv preprint* (2025). DOI: 10.48550/arXiv.2503.11926. arXiv: 2503.11926 [cs.AI].
- [6] OpenAI. *HealthBench: A Benchmark for Evaluating Clinical Conversations*. <https://huggingface.co/datasets/openai/healthbench>. Accessed 2025-12-10. 2025.
- [7] Yixing Jiang et al. “MedAgentBench: A Realistic Virtual EHR Environment to Benchmark Medical LLM Agents”. In: *arXiv preprint* (2025). DOI: 10.48550/arXiv.2501.14654. arXiv: 2501.14654 [cs.LG].

A Appendix Title

A.1 Standardized Clinical Reasoning Evaluation Rubric

This rubric was used to evaluate model-generated clinical reasoning along five dimensions. Each dimension has a definition and a 1-5 scale with descriptive anchors.

Reasoning Transparency

Definition: Assesses whether the medical rationale is explicitly stated, logically structured, and traceable.

Scoring (1-5):

1. No justification or contradictory.
2. Minimal or fragmented justification.
3. Basic logical sequence with at least one explicit connector.
4. Mostly structured and interpretable with minor omissions.
5. Fully explicit, clear, and stepwise reasoning.

Chain-of-Thought Quality

Definition: Evaluates clarity, completeness, and multi-step inferential progression.

Scoring (1-5):

1. No coherent chain or contradictory.
2. Single-step reasoning only.
3. Partial chain with two or more connected steps.
4. Mostly complete, coherent, and clinically meaningful.
5. Comprehensive, multi-step reasoning including rationale and alternatives.

Clinical Knowledge Application

Definition: Measures alignment with correct medical knowledge, standards, and guideline-supported decision making.

Scoring (1-5):

1. Unsafe or contraindicated recommendations.
2. Multiple significant knowledge errors.
3. Mostly correct but missing key criteria.
4. Correct and guideline-aligned with minor refinements missing.
5. Fully correct, guideline-based, and clinically optimal.

Contextual Understanding

Definition: Evaluates incorporation of patient-specific data, comorbidities, risks, and situational factors.

Scoring (1-5):

1. Entirely generic.
2. Minimal tailoring.
3. Incorporates at least one relevant contextual factor.
4. Mostly personalized with only minor omissions.
5. Fully tailored and integrates multiple relevant clinical and contextual elements.

Hallucination Audit

Definition: Scores factual grounding strictly against the patient note. Higher score indicates fewer unsupported claims.

Scoring (1-5):

1. Critical hallucinations with safety risk.
2. Major unsupported claims.
3. Multiple unsupported elements with moderate impact.
4. Minor unsupported detail with low impact.
5. No hallucination and fully supported or inferable.

A.2 ICD Code Lists

A.2.1 Head and Neck Cancer ICD-9 Diagnosis Codes

ICD-9-CM Code	Diagnosis	Description
141.0		Malignant neoplasm of base of tongue
141.1		Malignant neoplasm of dorsal surface of tongue
145.3		Malignant neoplasm of soft palate
146.0		Malignant neoplasm of tonsil
146.2		Malignant neoplasm of tonsillar pillars (anterior, posterior)
146.3		Malignant neoplasm of vallecula epiglottica
146.4		Malignant neoplasm of anterior aspect of epiglottis
146.5		Malignant neoplasm of junctional region of oropharynx
146.6		Malignant neoplasm of lateral wall of oropharynx
146.9		Malignant neoplasm of oropharynx, unspecified site

Table A.1: Head and neck cancer ICD-9 diagnosis codes.

A.2.2 Inguinal Hernia Repair ICD-9 Procedure Codes

ICD-9-CM Code	Procedure	Description
53.00		Unilateral repair of inguinal hernia, not otherwise specified
53.01		Other and open repair of direct inguinal hernia
53.02		Other and open repair of indirect inguinal hernia
53.03		Other and open repair of direct inguinal hernia with graft or prosthesis
53.04		Other and open repair of indirect inguinal hernia with graft or prosthesis
53.05		Repair of inguinal hernia with graft or prosthesis, not otherwise specified

Table A.2: Hippocrates-Chirurgos-01 (“SurgiMind”): inguinal hernia repair ICD-9 procedure codes.

A.3 Sample Clinical Vignettes

Discipline	Clinical Vignette
Bariatric	Laparoscopic Roux-en-Y gastric bypass with Nissen fundoplication takedown and paraesophageal hernia repair. Performed for refractory GERD and morbid obesity with failed medical management.
Bariatric	Laparoscopic Roux-en-Y gastric bypass. Performed for class III obesity (BMI 42.8) with insulin-dependent diabetes and nonalcoholic steatohepatitis, complicated by postoperative hemorrhage requiring re-exploration.
Surgical Oncology	Excisional biopsy of right neck mass. Diagnosed granulocytic sarcoma, acute myeloid leukemia confirmed on marrow. Started 7+3 induction chemo.
Surgical Oncology	VATS right upper lobe wedge resection with mediastinal lymph node dissection for enlarging PET-avid nodule. Pathology: 1.4 cm moderately differentiated squamous cell carcinoma, pT1aN0Mx, stage IA.
Aortic	Open aortic valve replacement and ascending aortic graft repair for severe bicuspid aortic stenosis with aneurysm.
Aortic	Emergency open repair of Type A aortic dissection with ascending aorta/total arch replacement and aortic valve repair.

Table A.3: Sample clinical vignettes for six illustrative patients.

A.4 Prompt Templates

A.4.1 Head and Neck Cancer Reasoning Task

System Prompt

You are a clinical reasoning evaluator. Read the discharge
↪ summary,
provided information and answer one oncology reasoning question.
↪ Do
not summarize the note. Perform classification plus
↪ justification.

User Prompt

Question:

Based on the discharge summary, what is the patients current
→ cancer
treatment phase and treatment intent? Treatment phase should be
→ one
of the following: active therapy, post treatment surveillance,
recurrent or metastatic management, or unclear. Treatment intent
should be one of the following: curative, palliative, or
→ unclear.
Provide a brief justification using evidence from the case.

Definitions: Active therapy means the patient is currently
→ receiving
cancer directed treatment. Post treatment surveillance means
→ prior
treatment is complete and the patient is being followed without
active therapy. Recurrent or metastatic management means there
→ is
recurrence or distant disease being actively managed. Curative
→ intent
means treatment is aimed at cure. Palliative intent means
→ treatment
is aimed at symptom control or life prolongation without cure.
→ If the
note does not have enough information, choose unclear and
→ explain why.

Output format:

```
{"treatment_phase": "<one of the four phase labels>",  
  "treatment_intent": "<one of the three intent labels>",  
  "justification": "<2 to 4 sentences citing evidence from the  
                  discharge summary>"}
```

Constraints:

Do not add extra fields. The answer and justification must be
→ based
only on the provided content.

Discharge summary:

<DISCHARGE_SUMMARY_TEXT>

Output JSON Schema Example

```
{ "treatment_phase": "active therapy",  
  "treatment_intent": "curative",  
  "justification": "The patient is currently receiving adjuvant
```

```
chemoradiotherapy after definitive resection of a
↳ locoregionally
confined head and neck squamous cell carcinoma, indicating
↳ ongoing
active cancer-directed treatment with curative intent."}
```

A.4.2 Inguinal Hernia Repair Reasoning Task

System Prompt

You are a precise clinical reasoning assistant.

User Prompt

You are a helpful medical assistant. Please analyze the
↳ following
patient case and provide your output in JSON format with exactly
↳ four
fields: "prompted_cot", "summary_surgery",
↳ "alternative_surgery", and
"clinical_reasoning".

1. "prompted_cot" should contain a step-by-step chain-of-thought reasoning that explains your thought process for all three questions.
2. "summary_surgery" should be a single sentence summarizing the surgery performed.
3. "alternative_surgery" should be a single sentence describing
↳ an
alternative surgery that could have been performed.
4. "clinical_reasoning" should be a single sentence explaining
↳ why
the patient received the specified surgery instead of the
alternative.

Do not include anything outside this JSON structure. Format your response exactly as JSON.

Patient case:
<DISCHARGE_SUMMARY_TEXT>

Output JSON Schema Example

```
{
  "prompted_cot": "Step 1: Identify the primary surgical problem
(e.g., recurrent ventral hernia). Step 2: Extract the
↳ operative
```



```

description to determine the exact procedure performed. Step
↳ 3:
Enumerate reasonable alternative surgical options for this
indication. Step 4: Compare the chosen operation with the
alternative in terms of risk, feasibility, and expected
↳ benefit for
this patient. Step 5: Conclude why the documented operation is
↳ most
appropriate.",
"summary_surgery": "The patient underwent a laparoscopic
↳ ventral
hernia repair with mesh placement.",
"alternative_surgery": "An alternative approach would have
↳ been an
open ventral hernia repair without laparoscopy.",
"clinical_reasoning": "Laparoscopic repair was favored over an
↳ open
approach to reduce wound complications and facilitate faster
recovery while still providing durable reinforcement with
↳ mesh."
}

```

A.4.3 Cross-Specialty Prompts

Bariatric Prompt

Bariatric Prompt:

Please provide 3 numbered sentences based on the patient case:

1. Summarize the surgery performed.
2. What alternative surgery could have been performed?
3. Explain the clinical reasoning for the patient receiving the
 - ↳ surgery specified, instead of the alternative?

Head/Neck Prompt

Head/Neck Prompt:

Please provide four sentences based on the patient case:

1. Summarize the surgery performed.
2. Explain the clinical reasoning for the patient receiving
 - ↳ postoperative chemotherapy, if applicable.
3. Explain the clinical reasoning for the patient receiving
 - ↳ postoperative radiotherapy, if applicable.
4. Explain the clinical reasoning for the patient receiving
 - ↳ postoperative immunotherapy, if applicable.

Aortic Prompt

Aortic Prompt:

Please provide three numbered sentences based on the patient

↪ case:

1. Summarize the surgery performed and indicate whether it was
↪ open or endovascular.
2. Identify the alternative approach (endovascular if open was
↪ performed, or open if endovascular was performed).
3. Explain the clinical rationale for selecting the chosen
↪ approach over the alternative.